How Did We Get Here?

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Difficulty: medium
Time Limit : 0.2 second

Description

We define the Power Modulo Sum of an array as follows:

1. Compute each element A[i] raised to the power of itself.
2. Take the remainder when divided by i+2, where i is the zero-based index of the element.
3. Sum these values.
4. Divide the total sum by n+1 (where n is the length of the array) and take the remainder.

Formally, for an array A of size n:

$$\text{Power Modulo Sum}(A) = \left(\sum_{i=0}^{n-1} (A[i]^{A[i]} \bmod (i+2)) \right) \bmod (n+1)$$

Example of the Power Modulo Sum Calculation

Let's compute the Power Modulo Sum of [3, 4, 5, 6]:

(3^3 MOD 2) = 27 MOD 2 = 1
(4^4 MOD 3) = 256 MOD 3 = 1

(5^5 MOD 4) = 3125 MOD 4 = 1
(6^6 MOD 5) = 46656 MOD 5 = 1
Now, sum the values:
1 + 1 + 1 + 1 = 4

Finally, take the remainder when divided by n+1 = 4+1 = 5:

4 MOD 5 = 4

Thus, Power Modulo Sum([3,4,5,6]) = 4.

Objectif

Given an array of integers, compute the sum of the Power Modulo Sums for every non-empty subset of the array.

Examples

Example 01
Input : 1 3 3
Subsets and their Power Modulo Sums:
[1] → (1^1 MOD 2) MOD 2 = 1 MOD 2 = 1
[3] → (3^3 MOD 2) MOD 2 = 1 MOD 2 = 1
[3] → (3^3 MOD 2) MOD 2 = 1 MOD 2 = 1
[1,3] → ((1^1 MOD 2) + (3^3 MOD 3)) MOD 3 = (1 + 0) MOD 3 = 1
[1,3] → ((1^1 MOD 2) + (3^3 MOD 3)) MOD 3 = (1 + 0) MOD 3 = 1
[3,3] → ((3^3 MOD 2) + (3^3 MOD 3)) MOD 3 = (1 + 0) MOD 3 = 1
[1,3,3] → ((1^1 MOD 2) + (3^3 MOD 3) + (3^3 MOD 4)) MOD 4 = (1 + 0 + 3) MOD 4 = 0
Output : 6

Example 02
Input: 2 5

Subsets and their Power Modulo Sums:

[2] → (2^2 MOD 2) MOD 2 = 0 MOD 2 = 0
[5] → (5^5 MOD 2) MOD 2 = 1 MOD 2 = 1
[2,5] → ((2^2 MOD 2) + (5^5 MOD 3)) MOD 3 = (0 + 2) MOD 3 = 2

Output: 3

Example 03
Input: 3 4 5 6 7 8

The sum of Power Modulo Sums for all subsets is 110.

Output: 110

Constraints
1 ≤ nums.length ≤ 20
1 ≤ nums[i] ≤ 10^9